

Class 4

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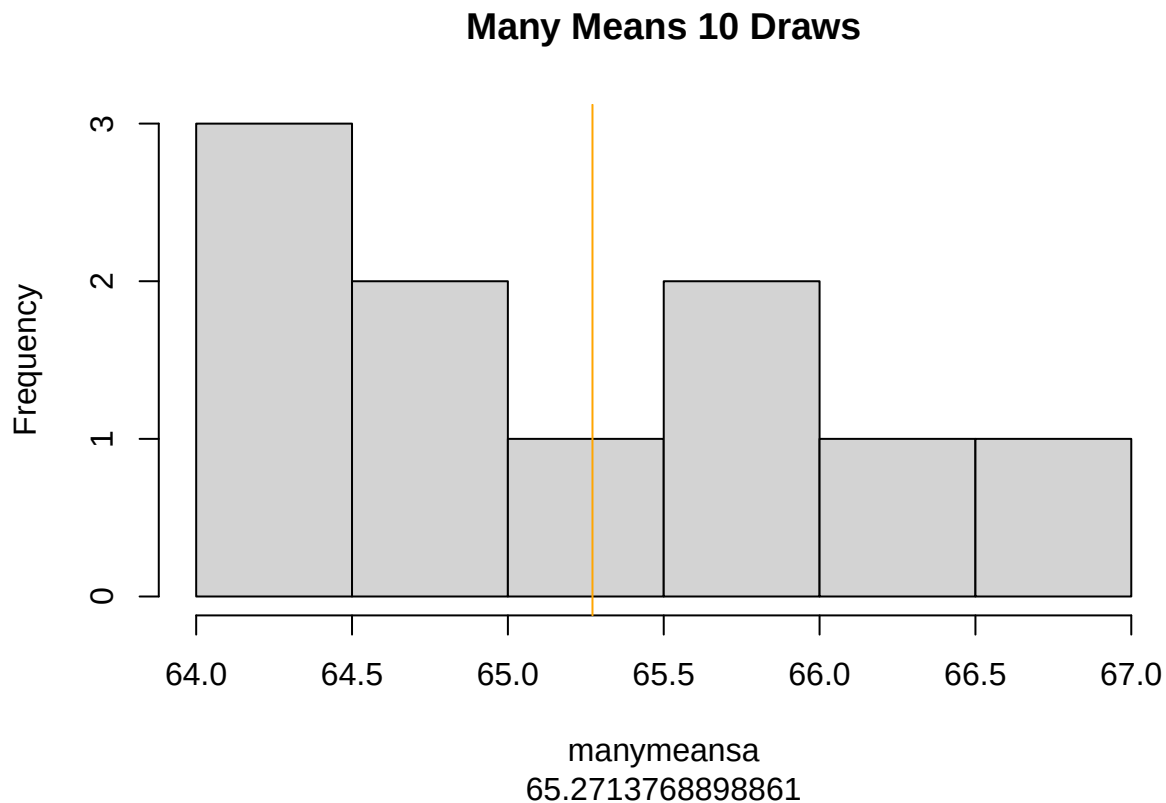
Distribution of the Mean

We'll start out illustrating concepts from discussion using R

```
getmean = function(N, mean, sd) {  
  opinion = rnorm(n = N, mean = mean, sd = sd)  
  meanopinion = mean(opinion)  
  return(meanopinion)  
}  
# Building a function in R. This is beyond the scope of the class  
  
manymeansa = replicate(n = 10, getmean(20, mean = 65, sd = 6))  
hist(manymeansa, main = "Many Means 10 Draws", sub = mean(manymeansa))  
mean(manymeansa)
```

```
## [1] 65.27138
```

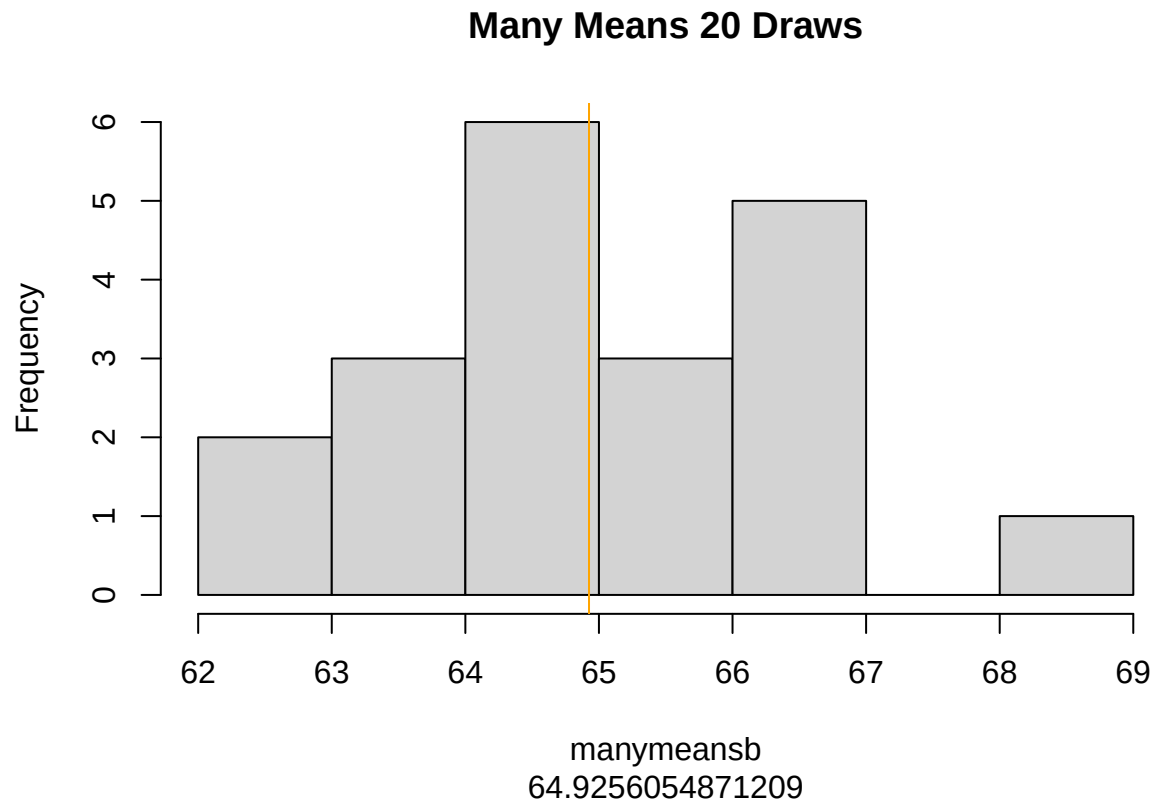
```
abline(v = mean(manymeansa), col = "orange")
```



```
# look with different n
manymeanb = replicate(n = 20, getmean(20, mean = 65, sd = 6))
hist(manymeanb, main = "Many Means 20 Draws", sub = mean(manymeanb))
mean(manymeanb)
```

```
## [1] 64.92561
```

```
abline(v = mean(manymeanb), col = "orange")
```

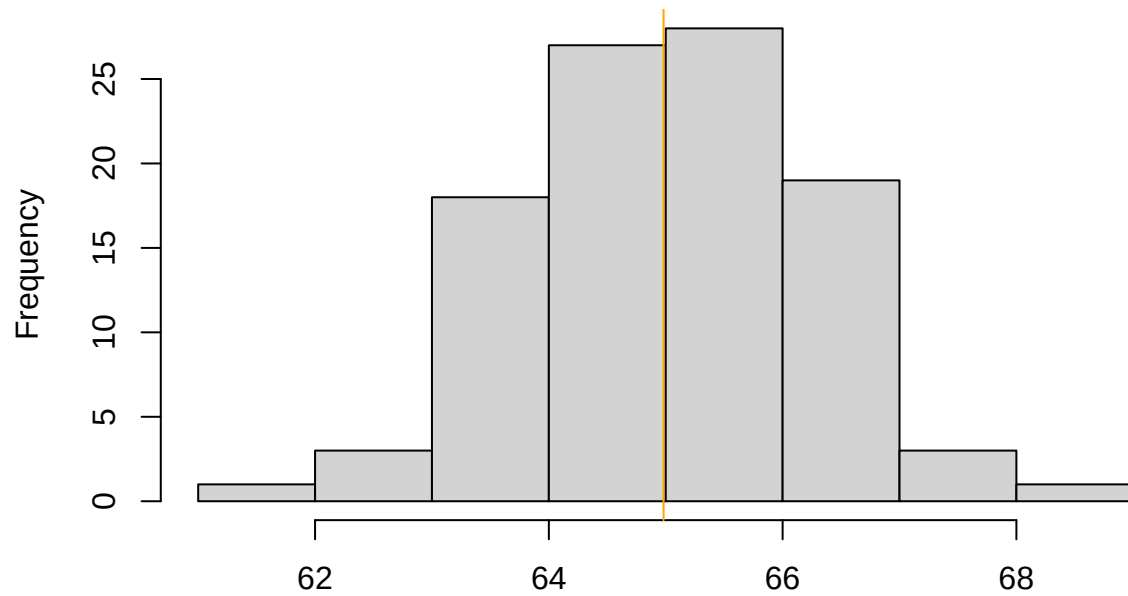


```
manymeanc = replicate(n = 100, getmean(20, mean = 65, sd = 6))
hist(manymeanc, main = "Many Means 100 Draws", sub = mean(manymeanc))
mean(manymeanc)
```

```
## [1] 64.98148
```

```
abline(v = mean(manymeanc), col = "orange")
```

Many Means 100 Draws



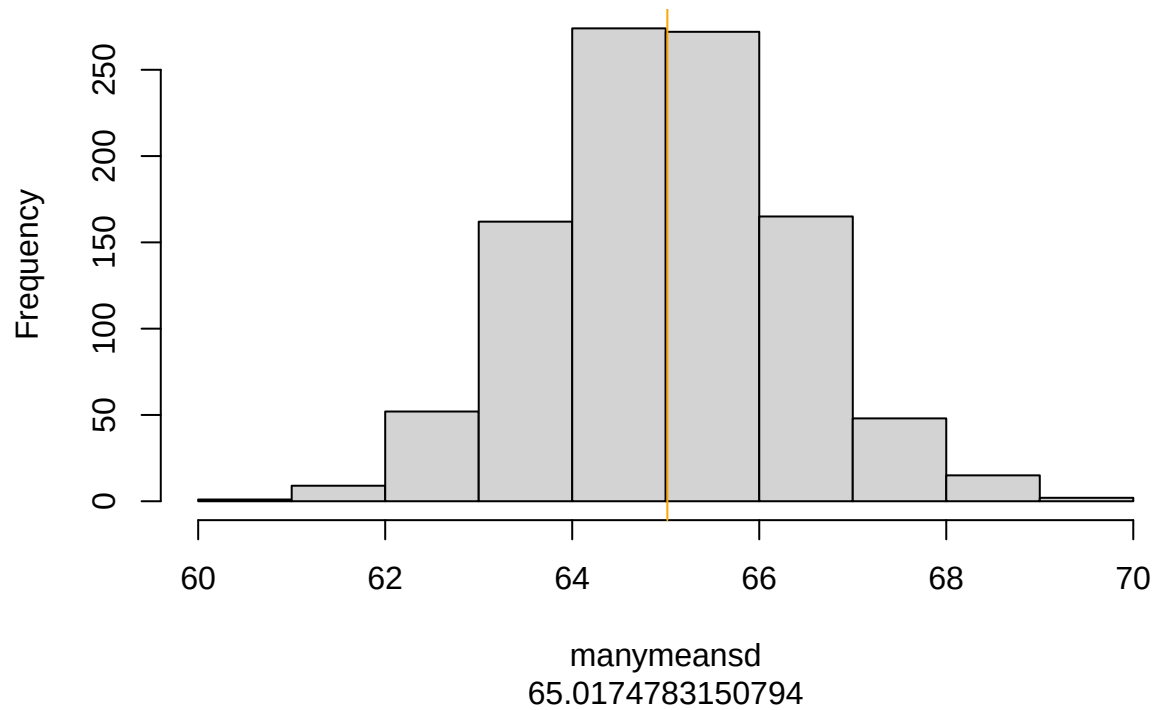
manymeansc
64.981477753395

```
manymeansd = replicate(n = 1000, getmean(20, mean = 65, sd = 6))  
hist(manymeansd, main = "Many Means 1k Draws", sub = mean(manymeansd))  
mean(manymeansd)
```

```
## [1] 65.01748
```

```
abline(v = mean(manymeansd), col = "orange")
```

Many Means 1k Draws

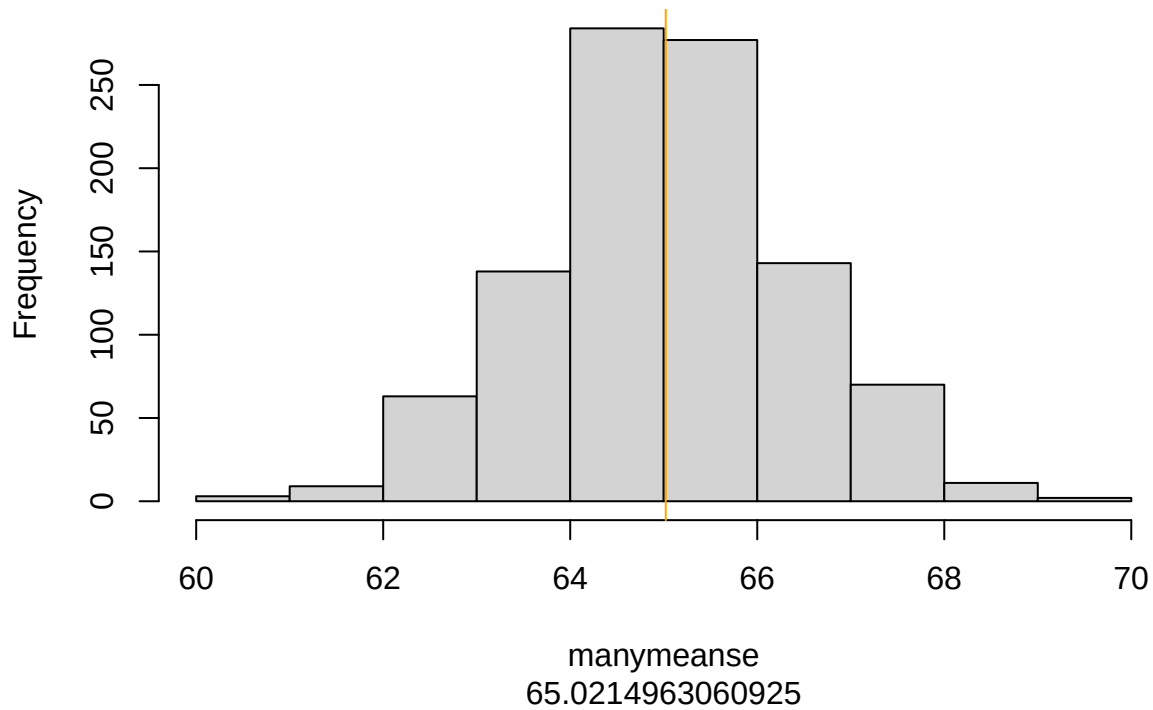


```
manymeanse = replicate(n = 1000, getmean(20, mean = 65, sd = 6))  
hist(manymeanse, main = "Many Means 1000 Draws", sub = mean(manymeanse))  
mean(manymeanse)
```

```
## [1] 65.0215
```

```
abline(v = mean(manymeanse), col = "orange")
```

Many Means 1000 Draws



```
manymeansf = replicate(n = 10000, getmean(20, mean = 65, sd = 6))  
hist(manymeansf, main = "Many Means 10k Draws", sub = mean(manymeansf))  
mean(manymeansf)
```

```
## [1] 65.00299
```

```
abline(v = mean(manymeansf), col = "orange")
```

```
# Larger sample sizes should be closer to 'true mean' but still variation
```

```
abline(v = 65, col = "blue")
```

```
abline(v = mean(manymeansa), col = "darkorange4")
```

```
abline(v = mean(manymeansb), col = "darkorange3")
```

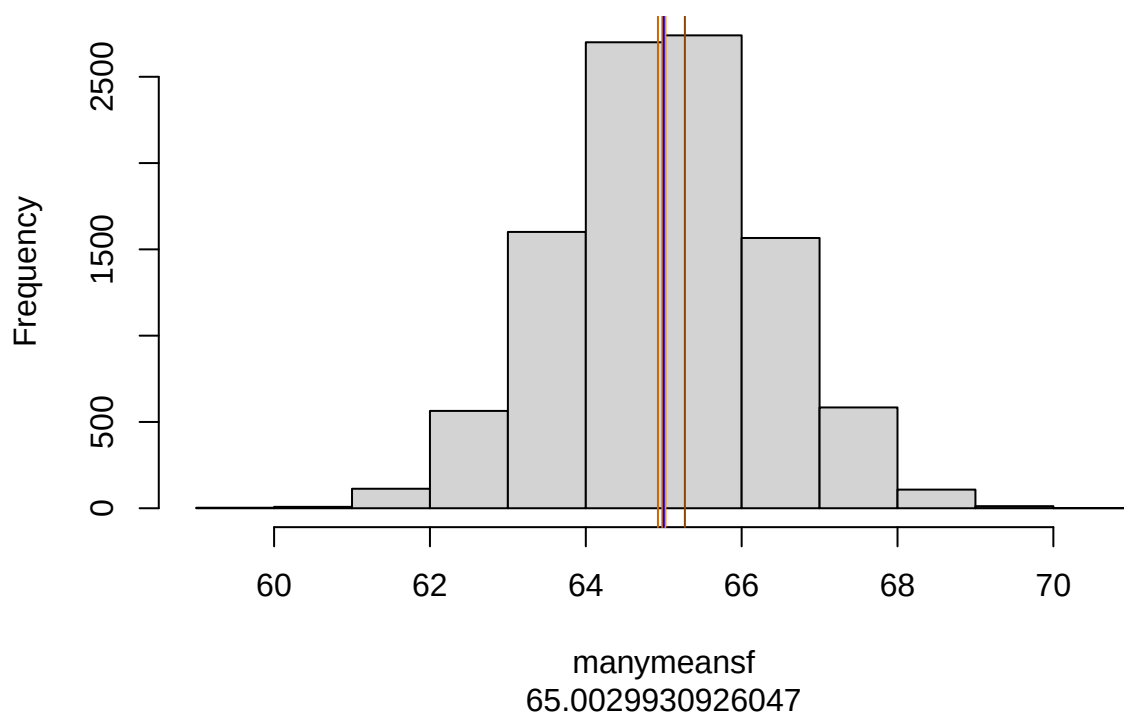
```
abline(v = mean(manymeansc), col = "darkorange2")
```

```
abline(v = mean(manymeansd), col = "darkorange1")
```

```
abline(v = mean(manymeanse), col = "darkorange")
```

```
abline(v = 65, col = "blue")
```

Many Means 10k Draws



```
mean(manymeansa)
```

```
## [1] 65.27138
```

```
mean(manymeansb)
```

```
## [1] 64.92561
```

```
mean(manymeansc)
```

```
## [1] 64.98148
```

```
mean(manymeansd)
```

```
## [1] 65.01748
```

```
mean(manymeanse)
```

```
## [1] 65.0215
```

```
mean(manymeansf) #as n increases will get closer to true mean
```

```
## [1] 65.00299
```

```
sd(manymeansa) # . . . and sd
```

```
## [1] 0.8426875
```

```
sd(manymeansb)
```

```
## [1] 1.45673
```

```
sd(manymeansc)
```

```
## [1] 1.258019
```

```
sd(manymeansd)

## [1] 1.322475

sd(manymeanse)

## [1] 1.376765

sd(manymeansf)

## [1] 1.339597
```

Setting the working directory(again)

Instead of using an Rproject this week we are getting to manually set our working directory (Translation: File where everything will be saved.) Always set your wd before working in R. This ensures that all your work(graphics, subsets, etc) will be saved in the same place and that R will know where to look for your data.

After the setwd() command insert the file path to your own folder that you will use for this class(If it doesn't exist create it using finder or in windows). Make sure both files I emailed you, this .rmd file and the .rdata file are in that folder.

There are two ways to have R tell you the file path. First, you can use the file.choose() to find the file path. Just copy and paste from the console into this code. The other is to load or set the wd directly in Rstudio, but if you do this you need paste into your code so it will knit.

This is perhaps the most counter-intuitive thing you'll do in R, but it is important. Remember, do not use point and click, you need the code to knit.

```
getwd() #Returns which directory you are currently working in

## [1] "/Users/colinkuehl/Dropbox/POLS 641/F26 641/f26course/Rcode/Week 4"

setwd("/Users/colinkuehl/Dropbox/POLS 641/F26 641/f26course/Rcode/Week 4") #Sets working directory - Y
getwd() #make sure using right folder

## [1] "/Users/colinkuehl/Dropbox/POLS 641/F26 641/f26course/Rcode/Week 4"
```

Facing Change Data

We're going to use data from the article "Facing Change: Gender and Climate Change Attitudes Worldwide" by Bush and Clayton we . Data can be found here: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/0Y8IKW>

This is just one of the many data sources they use.

```
pewdata <- read.csv("pewdata.csv") #If working directory is set correctly and this file is in it this
```

Lets explore the data: Variable names, # of variables, # of respondents

```
dim(pewdata)

## [1] 45435      18

names(pewdata)

## [1] "X"          "Country"    "Female"     "Ideo"       "ClimConcen"
## [6] "ClimSerious" "Tech"       "SupportParis" "WhenHarm"   "PesonalHarm"
## [11] "WorryType"   "RichPay"    "WomensRights" "Edu"        "StandIncome"
## [16] "ClimStand"   "GDPPC"      "Diffs"
```

Thats a lot of respondents.

```
table(pewdata$Country)
```

```
##
##           Argentina           Australia           Brazil
##           1000             1004             1000
##           Burkina Faso           Canada           Chile
##           1000             1004             1000
##           China           Ethiopia           France
##           3649             1000             1001
##           Germany           Ghana           India
##           1000             1000             2452
##           Indonesia           Israel           Italy
##           1000             1000             1000
##           Japan           Jordan           Kenya
##           1000             1000             1015
##           Lebanon           Malaysia           Mexico
##           1000             1000             1000
##           Nigeria           Pakistan Palestinian territories
##           1047             1200             1000
##           Peru           Philippines           Poland
##           1000             1000             1028
##           Russia           Senegal           South Africa
##           1002             1000             1000
##           South Korea           Spain           Tanzania
##           1005             1000             1000
##           Turkey           Uganda           Ukraine
##           947             1000             2079
##           United Kingdom           United States           Venezuela
##           999             1003             1000
##           Vietnam
##           1000
```

```
length(unique(pewdata$Country))
```

```
## [1] 40
```

```
length(unique(pewdata$WorryType))
```

```
## [1] 7
```

```
prop.table(table(pewdata$Country))
```

```
##
##           Argentina           Australia           Brazil
##           0.02200946           0.02209750           0.02200946
##           Burkina Faso           Canada           Chile
##           0.02200946           0.02209750           0.02200946
##           China           Ethiopia           France
##           0.08031253           0.02200946           0.02203147
##           Germany           Ghana           India
##           0.02200946           0.02200946           0.05396721
##           Indonesia           Israel           Italy
##           0.02200946           0.02200946           0.02200946
##           Japan           Jordan           Kenya
##           0.02200946           0.02200946           0.02233961
```

```
##           Lebanon           Malaysia           Mexico
##           0.02200946           0.02200946           0.02200946
##           Nigeria           Pakistan Palestinian territories
##           0.02304391           0.02641136           0.02200946
##           Peru           Philippines           Poland
##           0.02200946           0.02200946           0.02262573
##           Russia           Senegal           South Africa
##           0.02205348           0.02200946           0.02200946
##           South Korea           Spain           Tanzania
##           0.02211951           0.02200946           0.02200946
##           Turkey           Uganda           Ukraine
##           0.02084296           0.02200946           0.04575768
##           United Kingdom           United States           Venezuela
##           0.02198745           0.02207549           0.02200946
##           Vietnam
##           0.02200946
```

```
str(pewdata) #This tells us what type of data R thinks we have
```

```
## 'data.frame':   45435 obs. of  18 variables:
## $ X           : int  1 2 3 4 5 6 7 8 9 10 ...
## $ Country      : chr  "Argentina" "Argentina" "Argentina" "Argentina" ...
## $ Female       : int  0 0 1 0 1 0 1 1 0 1 ...
## $ Ideo         : int  5 3 4 4 4 5 NA 7 NA NA ...
## $ ClimConcen   : int  4 3 4 4 2 3 4 4 3 3 ...
## $ ClimSerious  : int  4 3 3 3 3 3 4 4 3 4 ...
## $ Tech         : int  0 0 0 0 0 1 1 0 1 1 ...
## $ SupportParis : int  1 1 0 0 0 NA 0 0 0 0 ...
## $ WhenHarm     : int  1 2 2 1 3 3 1 1 2 2 ...
## $ PesonalHarm  : int  4 3 4 4 2 2 4 4 1 3 ...
## $ WorryType    : chr  "Severe weather, like floods or intense storms" "Droughts or water shortages"
## $ RichPay      : int  0 1 1 1 1 NA 1 1 1 1 ...
## $ WomensRights : int  4 3 3 4 4 3 4 4 3 NA ...
## $ Edu          : int  7 11 7 9 16 6 7 6 7 7 ...
## $ StandIncome  : num -0.235 0 0 0 0 ...
## $ ClimStand    : num  0.806 -0.44 -0.44 -0.44 -0.44 ...
## $ GDPPC        : num  14398 14398 14398 14398 14398 ...
## $ Diffs        : num  -0.0105 -0.0105 -0.0105 -0.0105 -0.0105 ...
```

Look at climate concern

```
mean(pewdata$ClimConcen) #Doesn't know what to do with NAs
```

```
## [1] NA
```

```
mean(pewdata$ClimConcen, na.rm = T)
```

```
## [1] 3.246295
```

```
median(pewdata$ClimConce, na.rm = T)
```

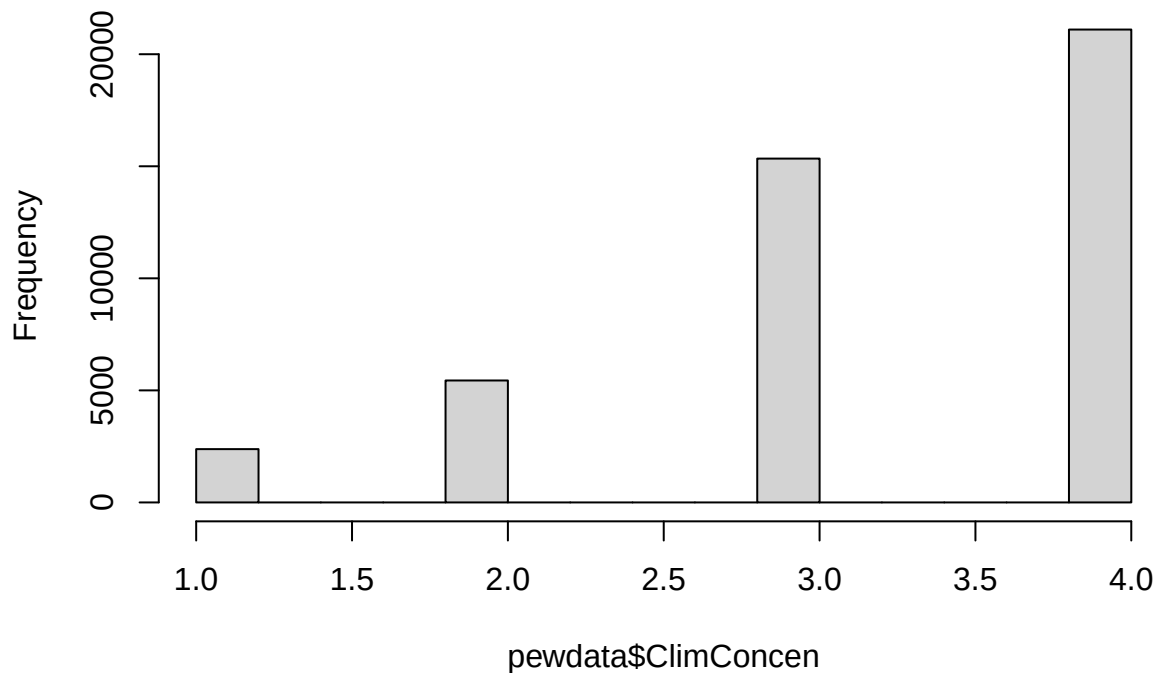
```
## [1] 3
```

```
summary(pewdata$ClimConcen)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##      1.000   3.000   3.000   3.246   4.000   4.000   1171
```

```
hist(pewdata$ClimConcen)
```

Histogram of pewdata\$ClimConcen



```
# A good way to count NAs in a column
sum(is.na(pewdata$ClimConcen))
```

```
## [1] 1171
```

```
sum(is.na(pewdata$Ideo))
```

```
## [1] 36172
```

```
table(pewdata$Country, pewdata$Ideo)
```

```
##
##           1   2   3   4   5   6   7
## Argentina 42  27  71 363  97  43  75
## Australia  0   0   0   0   0   0   0
## Brazil    45  66  73 314  86  69  75
## Burkina Faso 0   0   0   0   0   0   0
## Canada     0   0   0   0   0   0   0
## Chile     11  44 115 199  68  48  46
## China      0   0   0   0   0   0   0
## Ethiopia   0   0   0   0   0   0   0
## France     55  41 126 403 192 105  50
## Germany    42  50 150 481 159  35  20
## Ghana      0   0   0   0   0   0   0
## India       0   0   0   0   0   0   0
## Indonesia  0   0   0   0   0   0   0
## Israel      0   0   0   0   0   0   0
## Italy       17  57 148 237 137  88  37
## Japan       0   0   0   0   0   0   0
```

##	Jordan	0	0	0	0	0	0	0
##	Kenya	0	0	0	0	0	0	0
##	Lebanon	0	0	0	0	0	0	0
##	Malaysia	0	0	0	0	0	0	0
##	Mexico	34	35	78	236	138	65	64
##	Nigeria	0	0	0	0	0	0	0
##	Pakistan	0	0	0	0	0	0	0
##	Palestinian territories	0	0	0	0	0	0	0
##	Peru	38	27	43	354	108	92	93
##	Philippines	0	0	0	0	0	0	0
##	Poland	34	25	55	244	155	104	110
##	Russia	0	0	0	0	0	0	0
##	Senegal	0	0	0	0	0	0	0
##	South Africa	0	0	0	0	0	0	0
##	South Korea	0	0	0	0	0	0	0
##	Spain	71	43	151	428	130	59	47
##	Tanzania	0	0	0	0	0	0	0
##	Turkey	0	0	0	0	0	0	0
##	Uganda	0	0	0	0	0	0	0
##	Ukraine	0	0	0	0	0	0	0
##	United Kingdom	37	47	136	379	156	54	38
##	United States	0	0	0	0	0	0	0
##	Venezuela	60	79	55	193	106	106	149
##	Vietnam	0	0	0	0	0	0	0

Sex differences

```
males <- pewdata[which(pewdata$Female == 0), ]
females <- pewdata[which(pewdata$Female == 1), ]
```

```
summary(males$ClimConcen)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
##	1.000	3.000	3.000	3.214	4.000	4.000	492

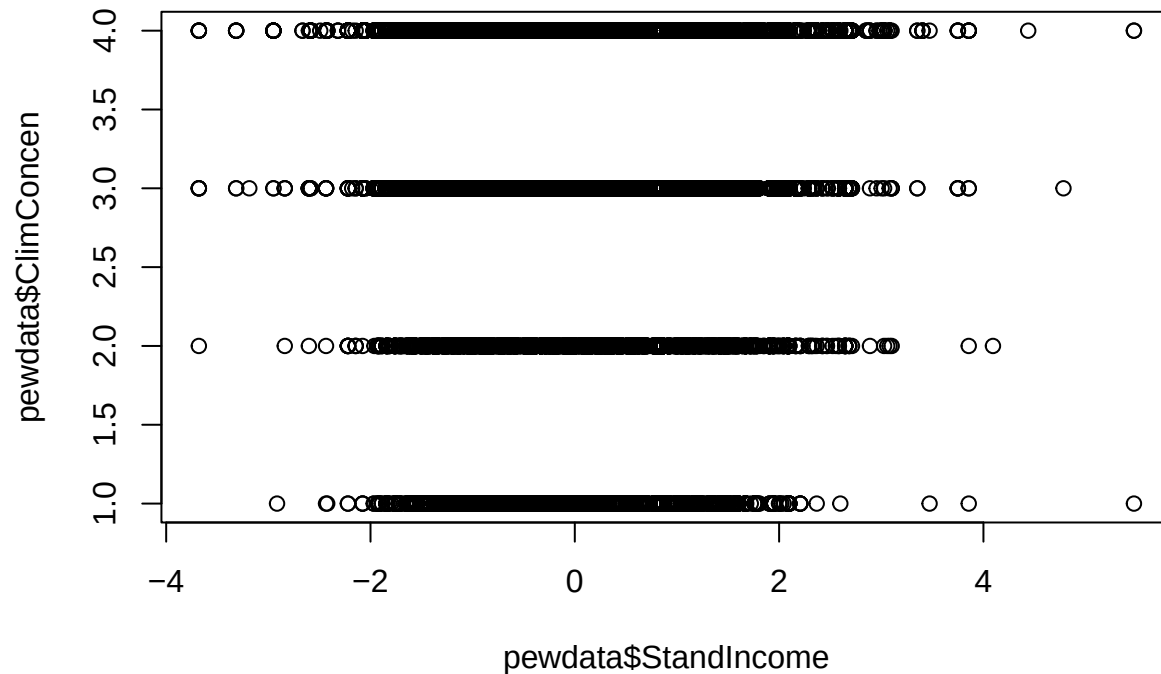
```
summary(females$ClimConcen)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
##	1.000	3.000	3.000	3.278	4.000	4.000	679

Lets say we we were interested in Income and climate concern

Use base R to create scatterplot

```
plot(pewdata$StandIncome, pewdata$ClimConcen) #Not very helpful
```



Reduce N to make more useful

```
pewsamp1 <- pewdata[sample(nrow(pewdata), 500), ] # a random sample of 500 ie a new subset
```

```
mean(pewsamp1$ClimConcen, na.rm = T)
```

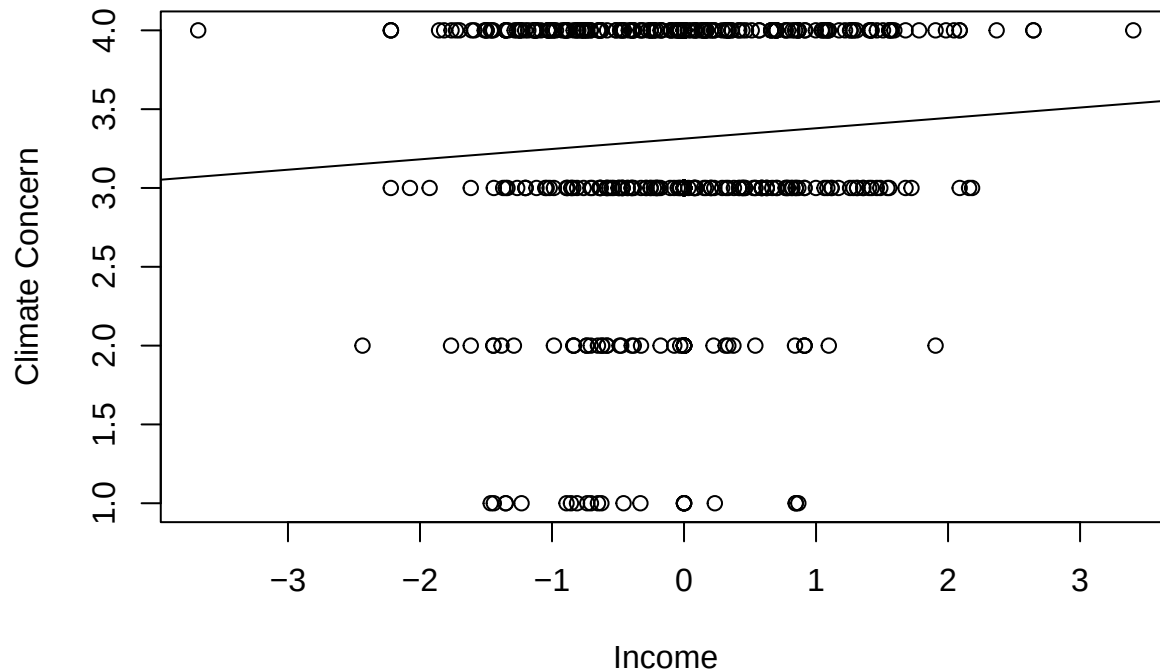
```
## [1] 3.308943
```

```
mean(pewdata$ClimConcen, na.rm = T)
```

```
## [1] 3.246295
```

```
plot(pewsamp1$StandIncome, pewsamp1$ClimConcen, main = "Income and Climate Concern",
      xlab = "Income", ylab = "Climate Concern") #our standard scatter plot
abline(lm(pewsamp1$ClimConcen ~ pewsamp1$StandIncome))
```

Income and Climate Concern



Using tidyverse

```
pewsamp2 <- sample_n(pewdata, 500) #Notice how tidyverse uses a slightly different vocabulary and grammar
```

```
mean(pewsamp2$ClimConcen, na.rm = T) #Ack! This is different from above
```

```
## [1] 3.203252
```

```
# We can create the same figure with the ggplot package. ggplot uses a different grammar.
```

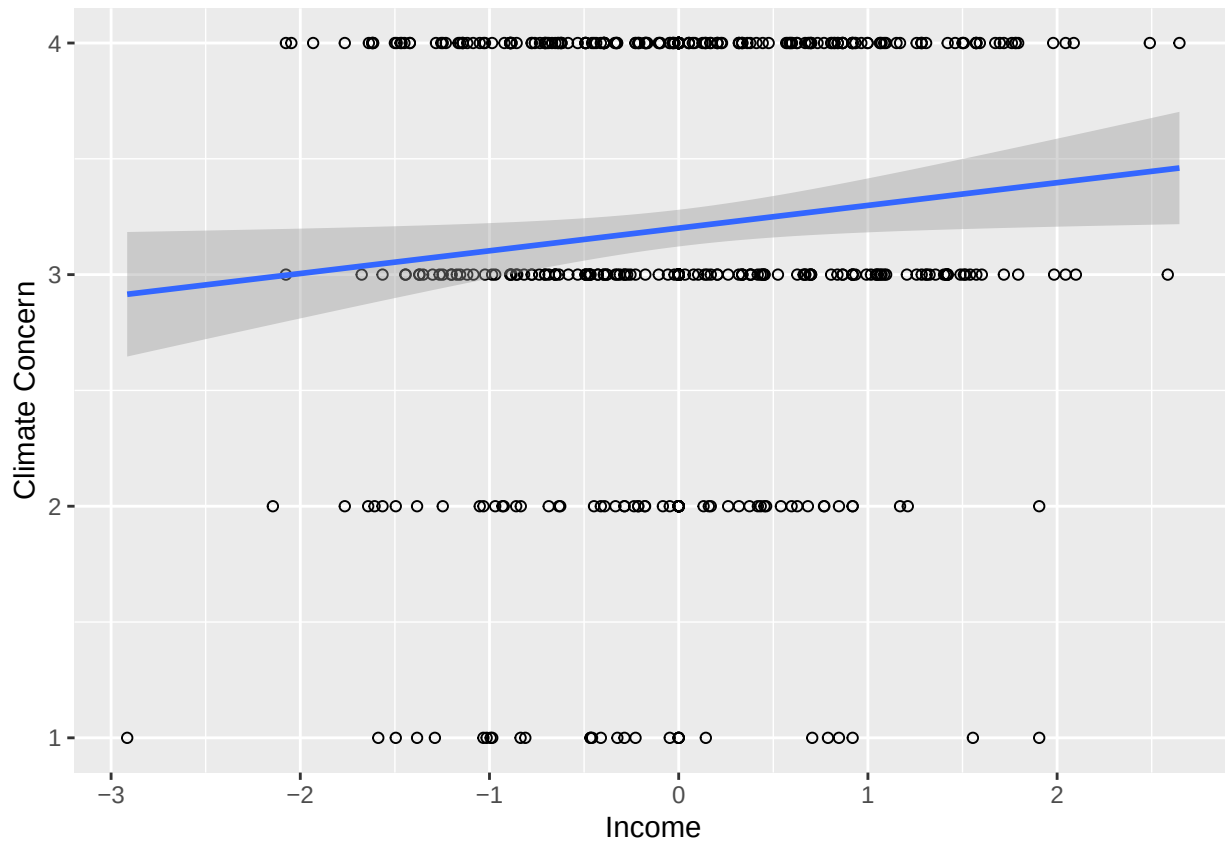
```
ggplot(data = pewsamp2, aes(x = StandIncome, y = ClimConcen)) + geom_point(shape = 1) +  
  xlab("Income") + ylab("Climate Concern") + geom_smooth(method = "lm", formula = y ~  
  x)
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
```

```
## (`stat_smooth()`).
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```



There are many ways to code in R. Base R can do everything you need, but I find “the tidyverse” more intuitive so I am introducing it early with the assumption you’ll find it easier as well.

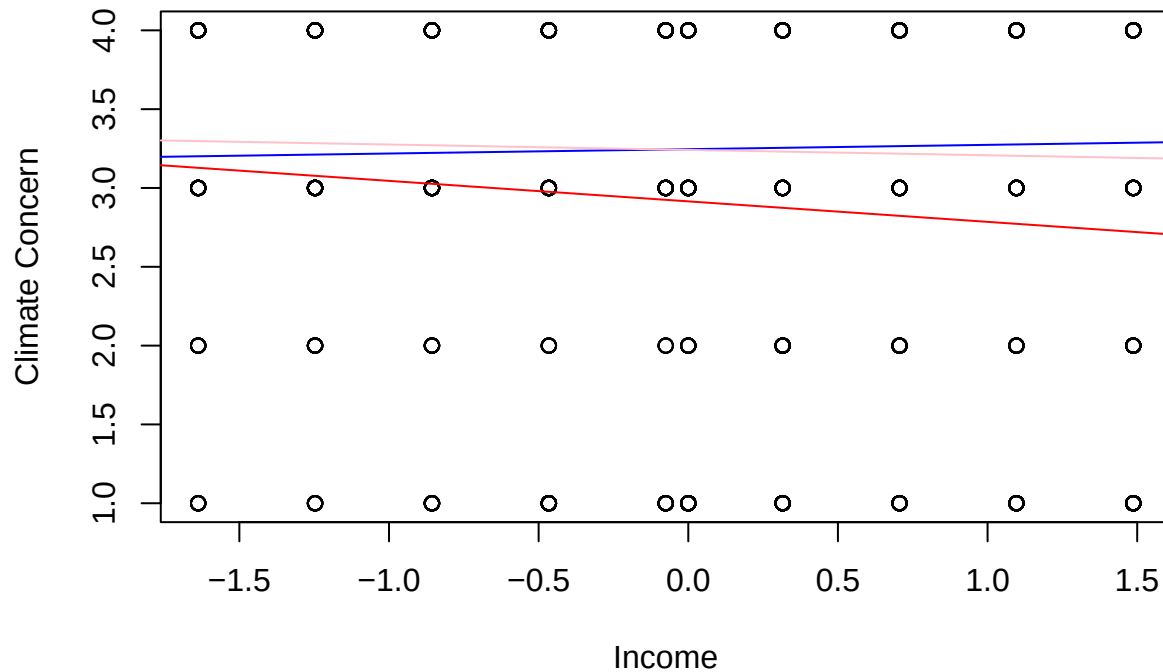
This is interesting, but might want to look at one country at a time.

Subset the data

```
usa <- pewdata[which(pewdata$Country == "United States"), ]
indo <- pewdata[which(pewdata$Country == "Indonesia"), ]

plot(usa$StandIncome, usa$ClimConcen, main = "Income and Climate Concern", xlab = "Income",
     ylab = "Climate Concern") #our standard scatter plot
abline(lm(usa$ClimConcen ~ usa$StandIncome), col = "red")
abline(lm(pewdata$ClimConcen ~ pewdata$StandIncome), col = "blue") #add line for whole sample to make
abline(lm(indo$ClimConcen ~ indo$StandIncome), col = "pink")
```

Income and Climate Concern



Create same graph using dplyr and ggplot

```
USA <- filter(pewdata, Country == "United States") #This is dplyr version of what we did last time. Ch
```

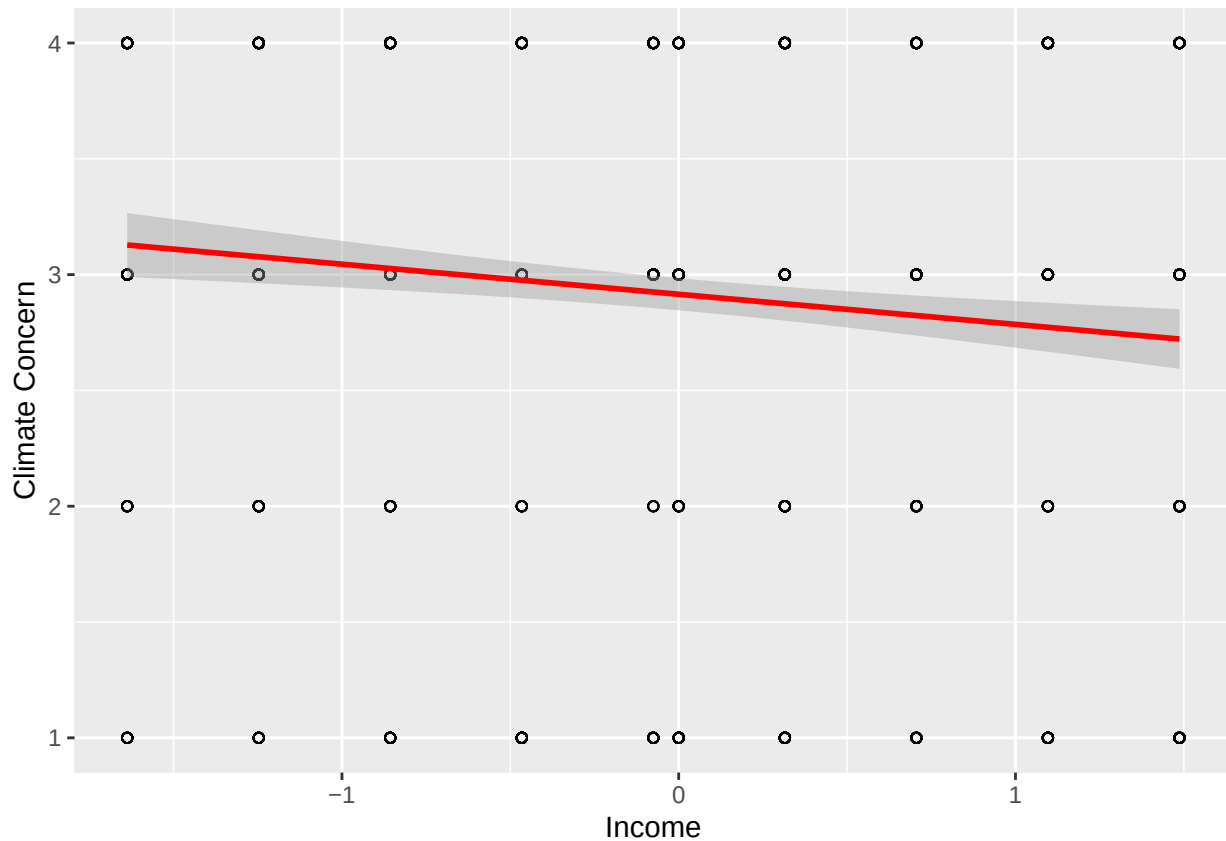
```
ggplot(USA, aes(x = StandIncome, y = ClimConcen)) + geom_point(shape = 1) + xlab("Income") +  
  ylab("Climate Concern") + geom_smooth(method = "lm", formula = y ~ x, col = "red")
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
```

```
## (`stat_smooth()`).
```

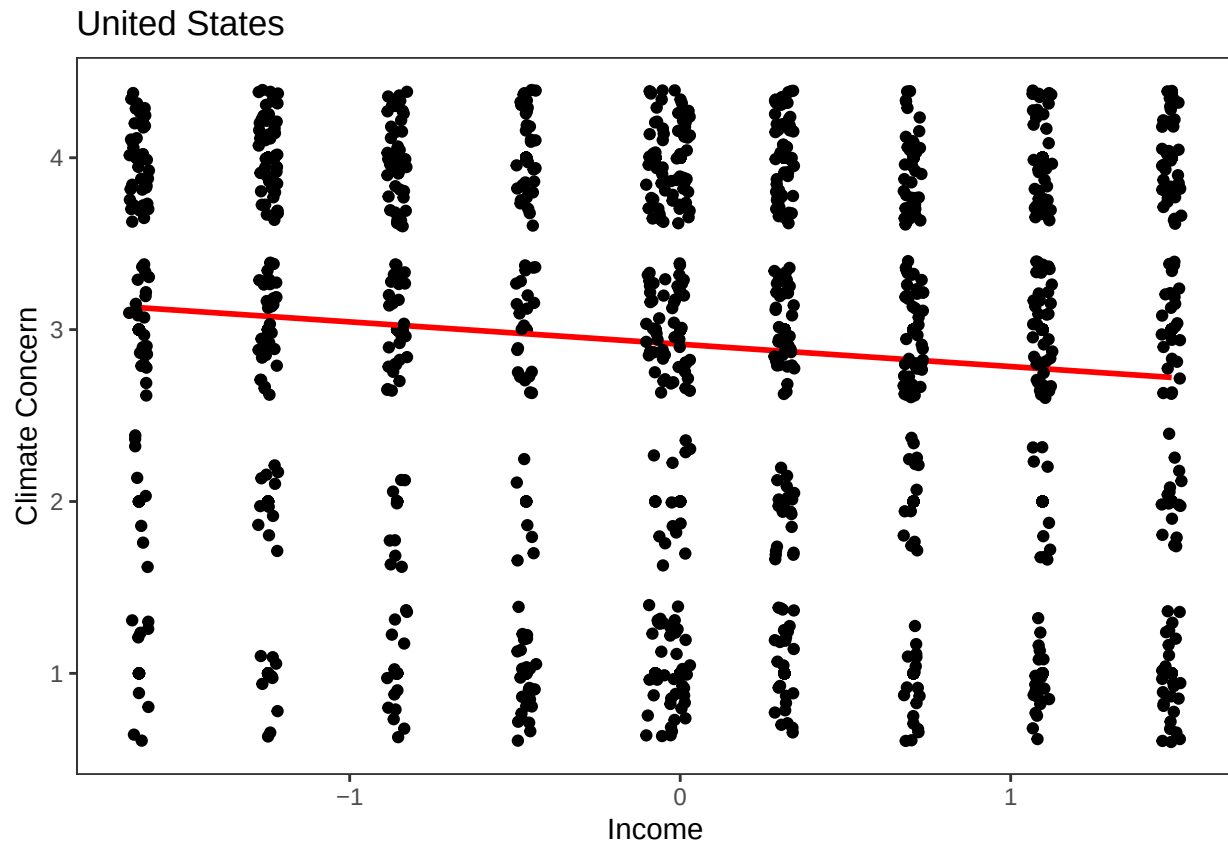
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```



```
ggplot(USA, aes(x = StandIncome, y = ClimConcen)) + geom_point() + ggtitle("United States") +
  xlab("Income") + ylab("Climate Concern") + geom_smooth(method = "lm", formula = y ~
    x, col = "red", se = F) + geom_jitter() + theme_test()
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range (`stat_smooth()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
## Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



An especially nice feature of tidy verse is that you can use piping(`%>%`) to do multiple things in a single line of code

```
pewdata %>%
  filter(Country == "Russia") %>%
  ggplot(aes(x = StandIncome, y = ClimConcen)) + geom_point() + ggtitle("Russia") +
  xlab("Income") + ylab("Climate Concern") + geom_smooth(method = "lm", formula = y ~
  x, col = "red") + geom_jitter() + theme_bw()
```

```
## Warning: Removed 19 rows containing non-finite outside the scale range
```

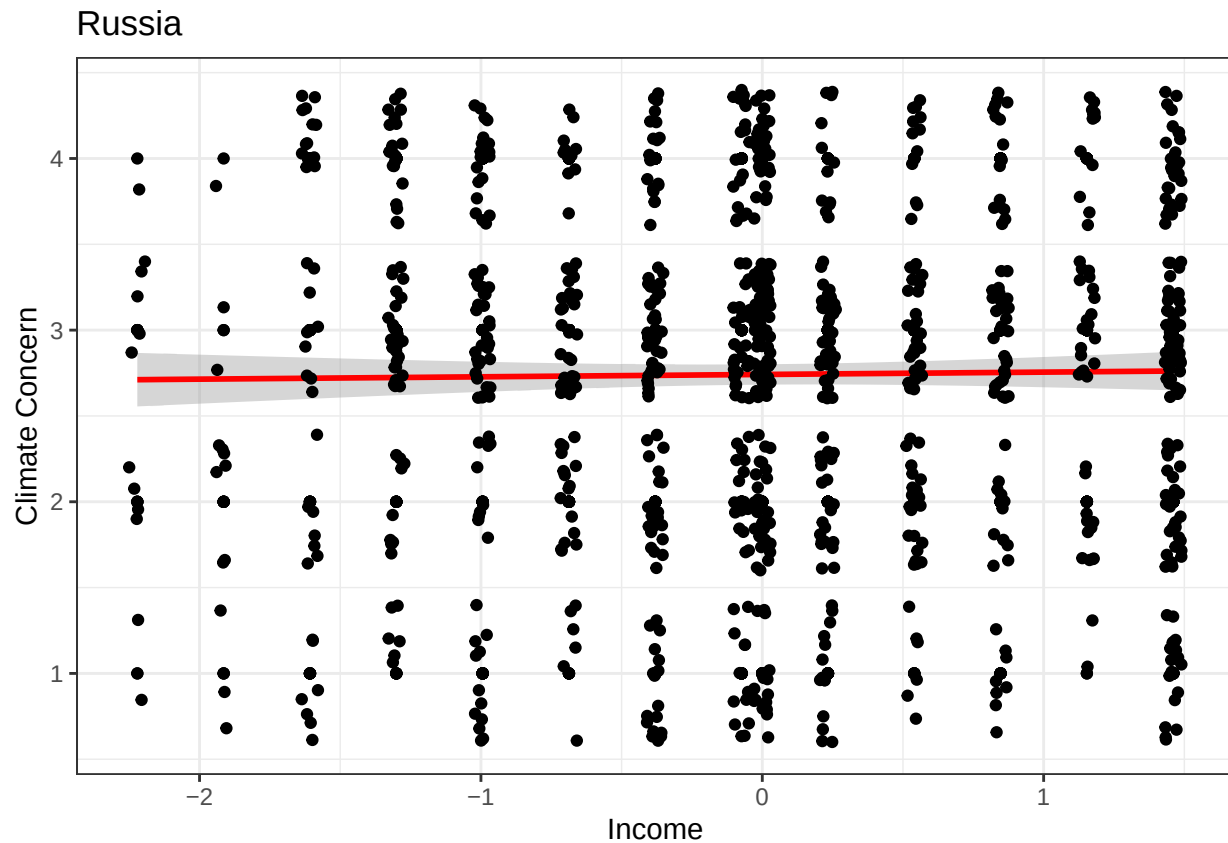
```
## (`stat_smooth()`).
```

```
## Warning: Removed 19 rows containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```

```
## Removed 19 rows containing missing values or values outside the scale range
```

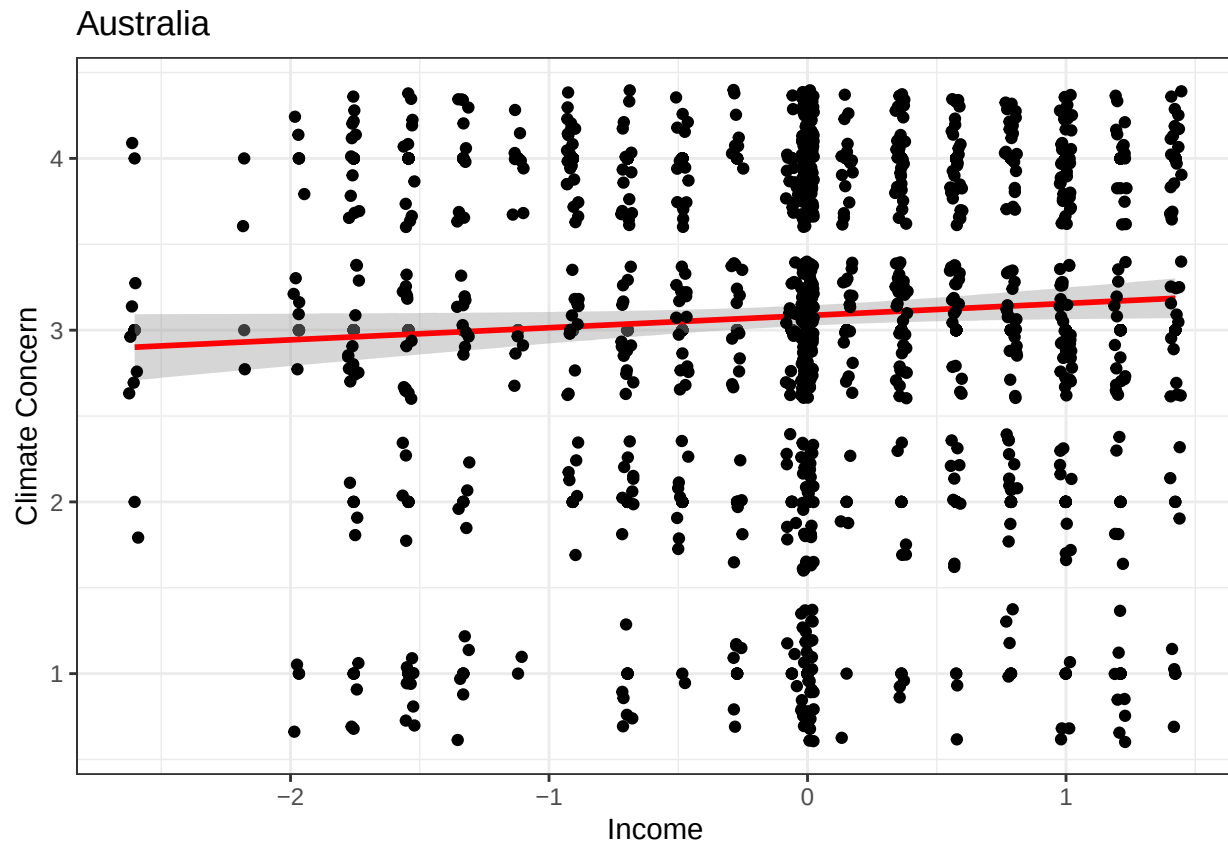
```
## (`geom_point()`).
```



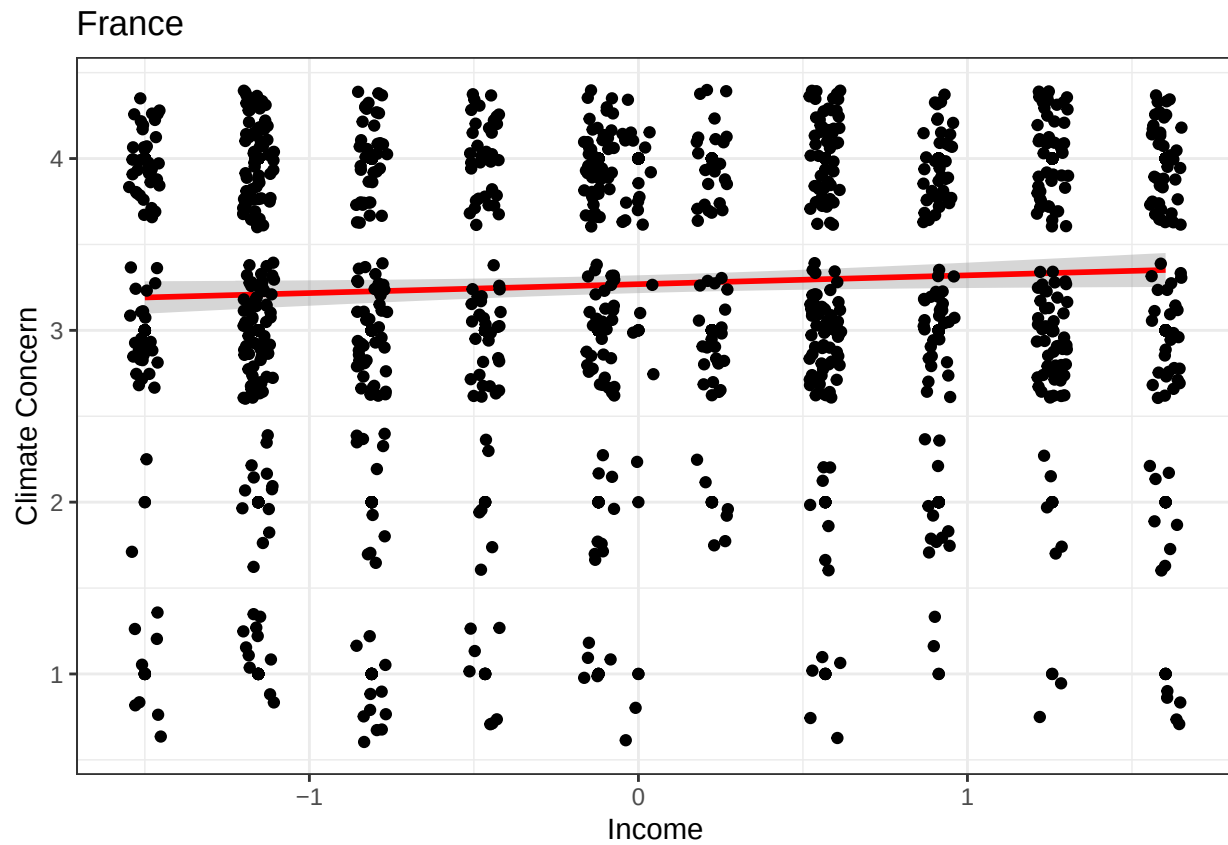
```
# This can make it much easier to run similar versions of the same graph
pewdata %>%
  filter(Country == "Australia") %>%
  ggplot(aes(x = StandIncome, y = ClimConcen)) + geom_point() + ggtitle("Australia") +
  xlab("Income") + ylab("Climate Concern") + geom_smooth(method = "lm", formula = y ~
  x, col = "red") + geom_jitter() + theme_bw()
```

```
## Warning: Removed 3 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

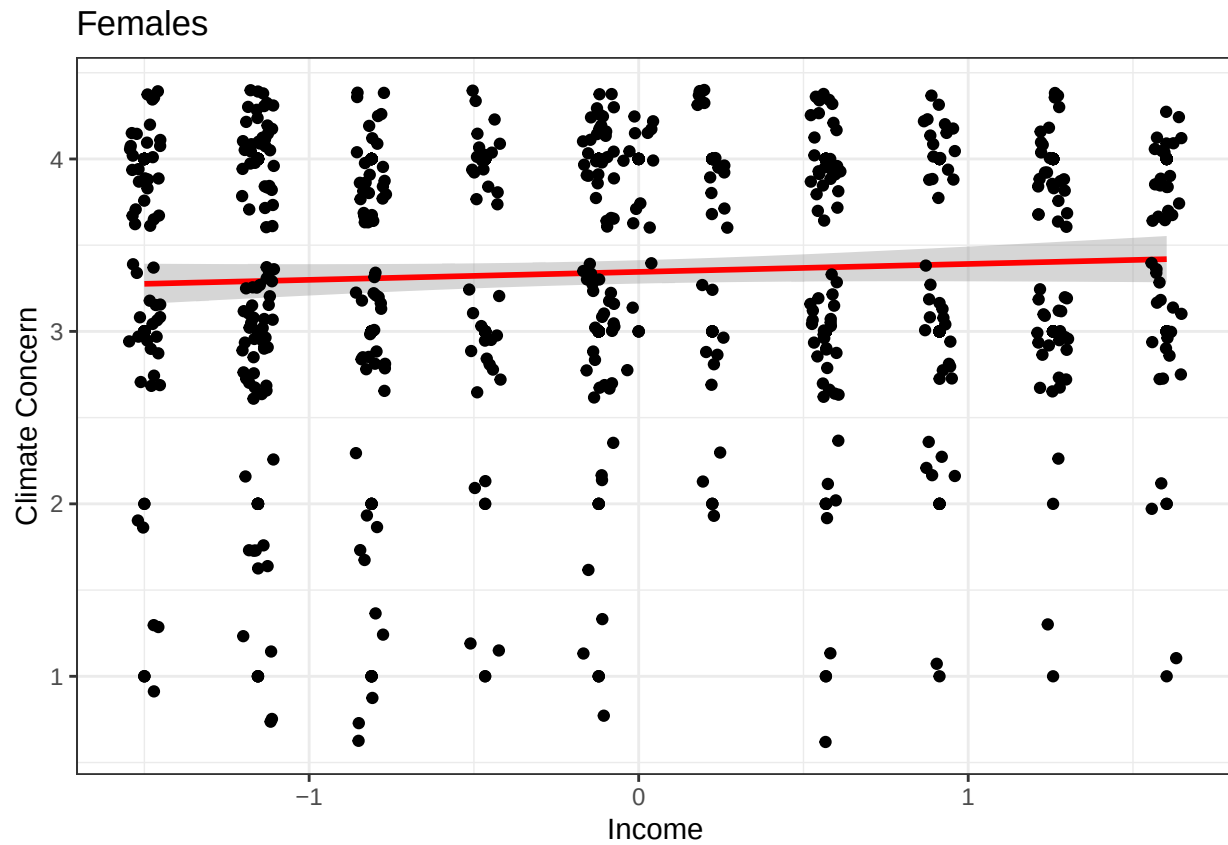
```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
## Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
pewdata %>%
  filter(Country == "France") %>%
  ggplot(aes(x = StandIncome, y = ClimConcen)) + geom_point() + ggtitle("France") +
  xlab("Income") + ylab("Climate Concern") + geom_smooth(method = "lm", formula = y ~
  x, col = "red") + geom_jitter() + theme_bw()
```



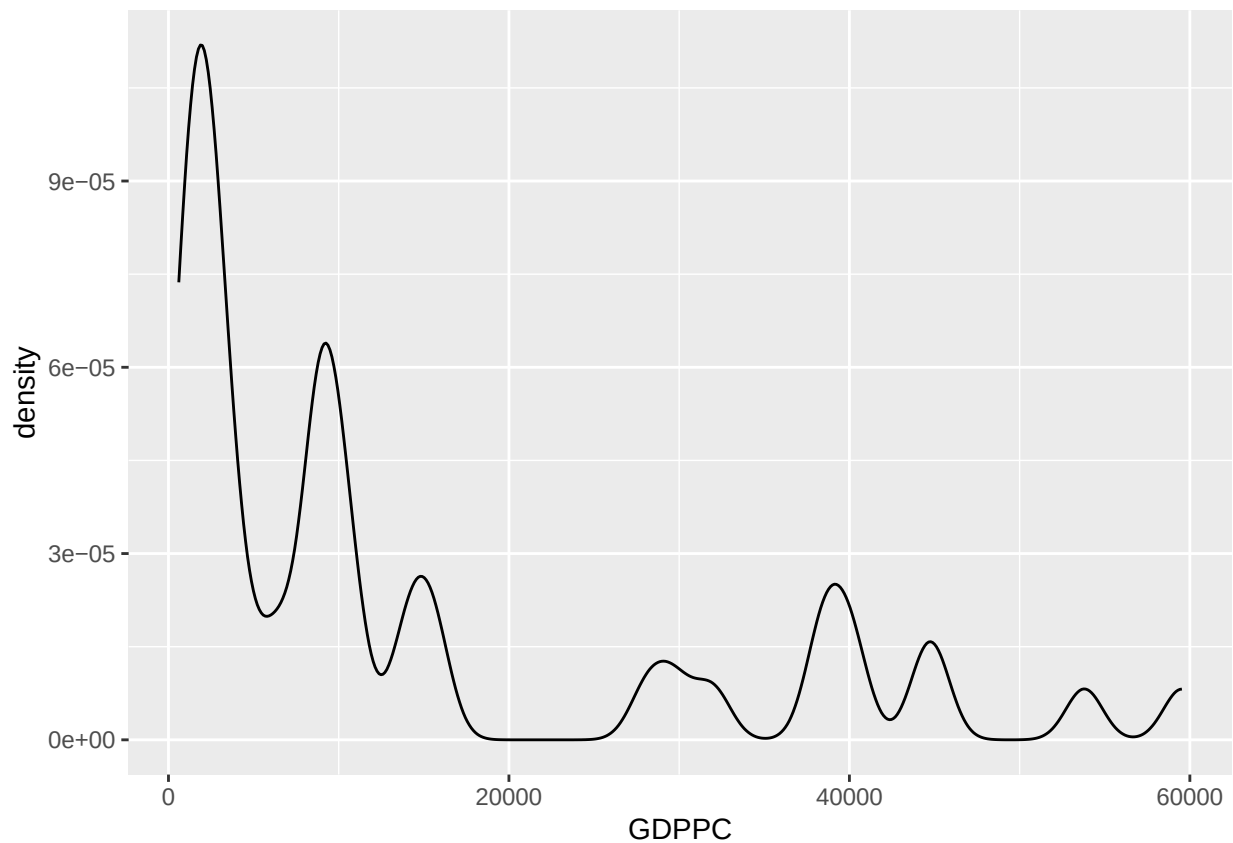
```
pewdata %>%
  filter(Female == 1, Country == "France") %>%
  ggplot(aes(x = StandIncome, y = ClimConcen)) + geom_point() + ggtitle("Females") +
  xlab("Income") + ylab("Climate Concern") + geom_smooth(method = "lm", formula = y ~
  x, col = "red") + geom_jitter() + theme_bw()
```



Your turn to play - create for another county, or look at another DV

Recreating Figure 1

```
ggplot(pewdata, aes(x = GDPPC)) + geom_density() #ggplot code for a histogram
```



```
pewdata$female <- ifelse(pewdata$Female == 1, "female", "male")
```

```
bycountryg <- pewdata %>%
  group_by(Country, female) %>%
  summarise(Avgseries = mean(ClimSeries, na.rm = T), GDP = mean(GDPPC)) %>%
  pivot_wider(names_from = female, values_from = Avgseries) %>%
  mutate(diff = female - male)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Country and female.
## i Output is grouped by Country.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Country, female))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

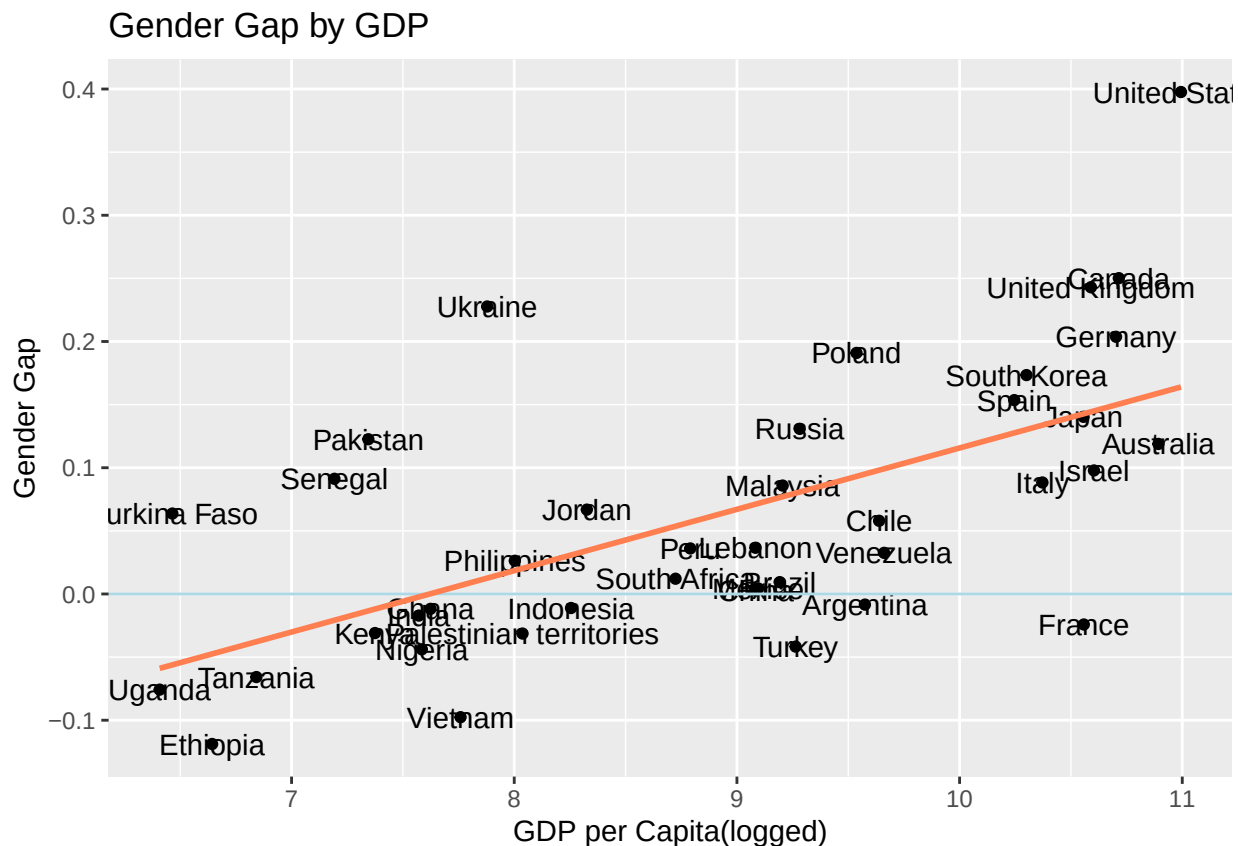
```
bycountryg
```

```
## # A tibble: 40 x 5
## # Groups:   Country [40]
##   Country      GDP female male    diff
##   <chr>      <dbl> <dbl> <dbl> <dbl>
## 1 Argentina  14398.   3.5  3.51 -0.00840
## 2 Australia  53800.   3.20  3.08  0.119
## 3 Brazil     9821.   3.84  3.83  0.00941
## 4 Burkina Faso  642    3.77  3.71  0.0637
## 5 Canada    45032.   3.42  3.17  0.250
## 6 Chile     15346.   3.79  3.73  0.0580
## 7 China      8827    2.96  2.95  0.00286
```

```
## 8 Ethiopia      768.    3.01  3.13 -0.119
## 9 France        38477.   3.44  3.47 -0.0242
## 10 Germany      44470.   3.49  3.29  0.204
## # i 30 more rows
```

```
ggplot(bycountryg, aes(x = log(GDP), y = diff, label = Country)) + geom_point() +
  geom_text() + ggtitle("Gender Gap by GDP") + xlab("GDP per Capita(logged)") +
  ylab("Gender Gap") + geom_hline(yintercept = 0, col = "lightblue") + geom_smooth(method = "lm",
  formula = y ~ x, col = "coral", se = F)
```

```
## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
## variable into a factor?
```



Ideology in Congress

As always Set your working and then Load the data(congress)

```
load("congress.RData")
```

We'll start by exploring the validity and reliability of the DW-Nominate Scale

This is a measure of members of congress ideology

- A Wikipedia entry is available [here](#).)
- A longer explainer is available [here](#).

- What do you think? (in terms of validity? reliability?)

Explore the data

```
head(congress)
```

```
##   congress district   state party      name dwnom1 dwnom2
## 1      80         0     USA Democrat  TRUMAN -0.276  0.016
## 2      80         1 ALABAMA Democrat  BOYKIN  F. -0.026  0.796
## 3      80         2 ALABAMA Democrat  GRANT   G. -0.042  0.999
## 4      80         3 ALABAMA Democrat  ANDREWS G. -0.008  1.005
## 5      80         4 ALABAMA Democrat  HOBBS   S. -0.082  1.066
## 6      80         5 ALABAMA Democrat  RAINS   A. -0.170  0.870
```

```
length(unique(congress$name))
```

```
## [1] 4082
```

```
## subset the data by party
```

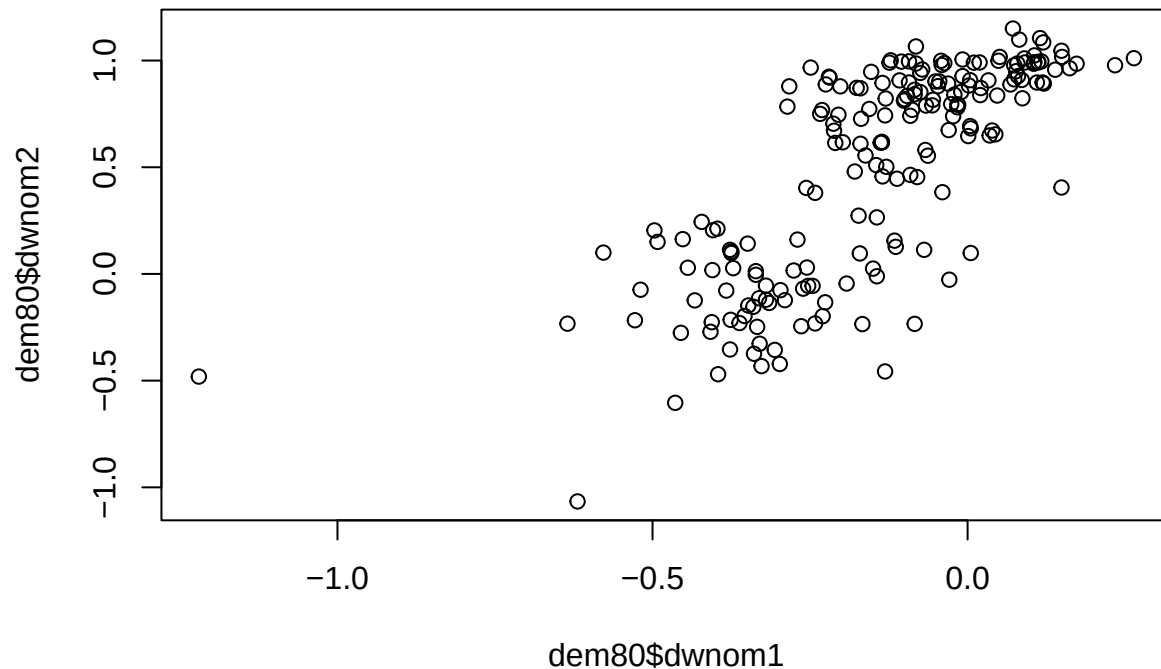
```
rep <- subset(congress, party == "Republican") #this command uses dplyr/tidyverse
dem <- congress[congress$party == "Democrat", ] #Uses base R, but does the same thing
```

Subset Further by Congress

```
rep80 <- subset(rep, subset = (congress == 80))
dem80 <- subset(dem, subset = (congress == 80))
rep112 <- subset(rep, subset = (congress == 112))
dem112 <- subset(dem, subset = (congress == 112))
```

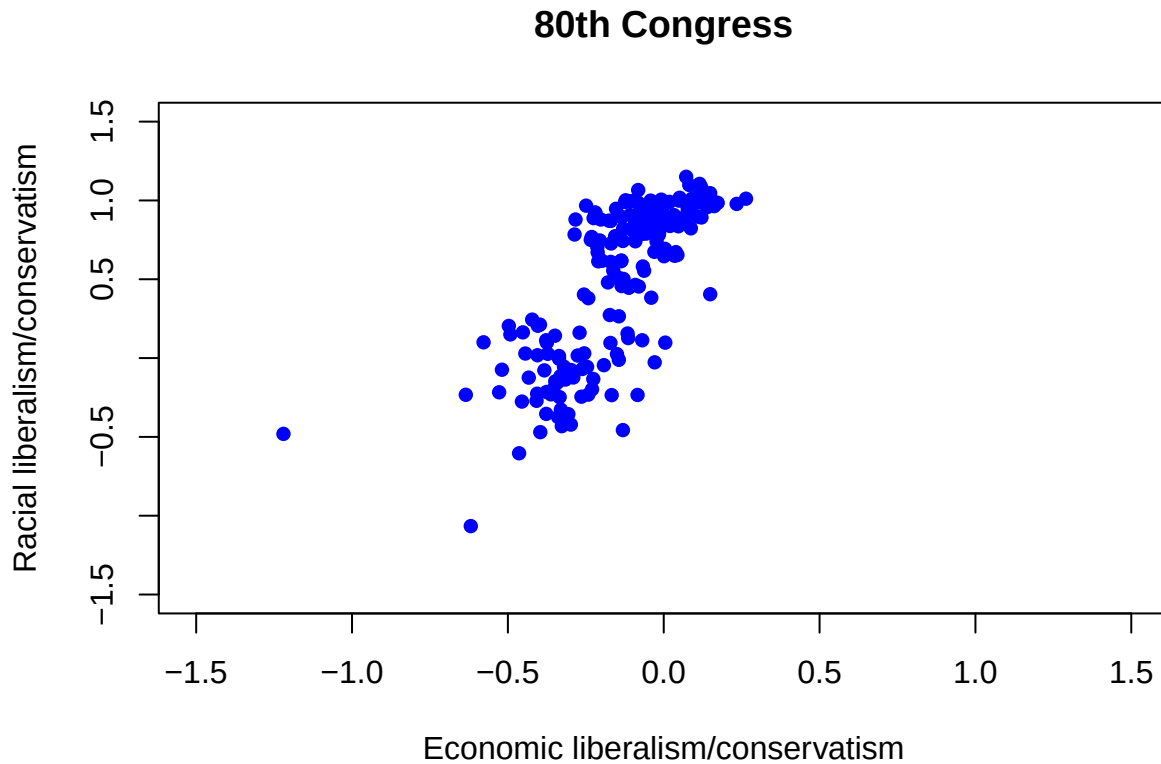
Make a Scatterplot

```
plot(dem80$dwnom1, dem80$dwnom2)
```



Make it Prettier

```
Demplot <- plot(dem80$dwnom1, dem80$dwnom2, pch = 16, col = "blue", xlim = c(-1.5, 1.5), ylim = c(-1.5, 1.5), xlab = "Economic liberalism/conservatism", ylab = "Racial liberalism/conservatism", main = "80th Congress")
```



What does this tell us about the two dimensions?

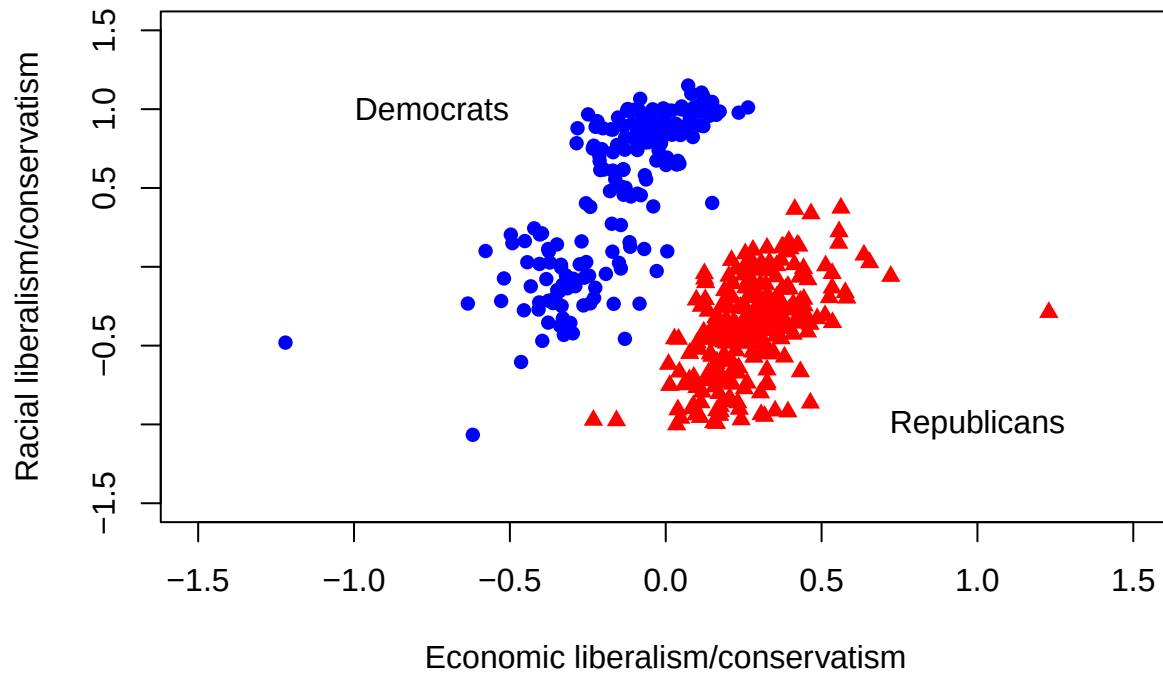
Let's add the GOP

```
plot(dem80$dwnom1, dem80$dwnom2, pch = 16, col = "blue", xlim = c(-1.5, 1.5), ylim = c(-1.5, 1.5), xlab = "Economic liberalism/conservatism", ylab = "Racial liberalism/conservatism", main = "80th Congress")
points(rep80$dwnom1, rep80$dwnom2, pch = 17, col = "red")
text(-0.75, 1, "Democrats") #adding a label to our plot
text(1, -1, "Republicans")
```

When plotting, you have to send ALL the commands at once

##Ain't it Perty!?

80th Congress

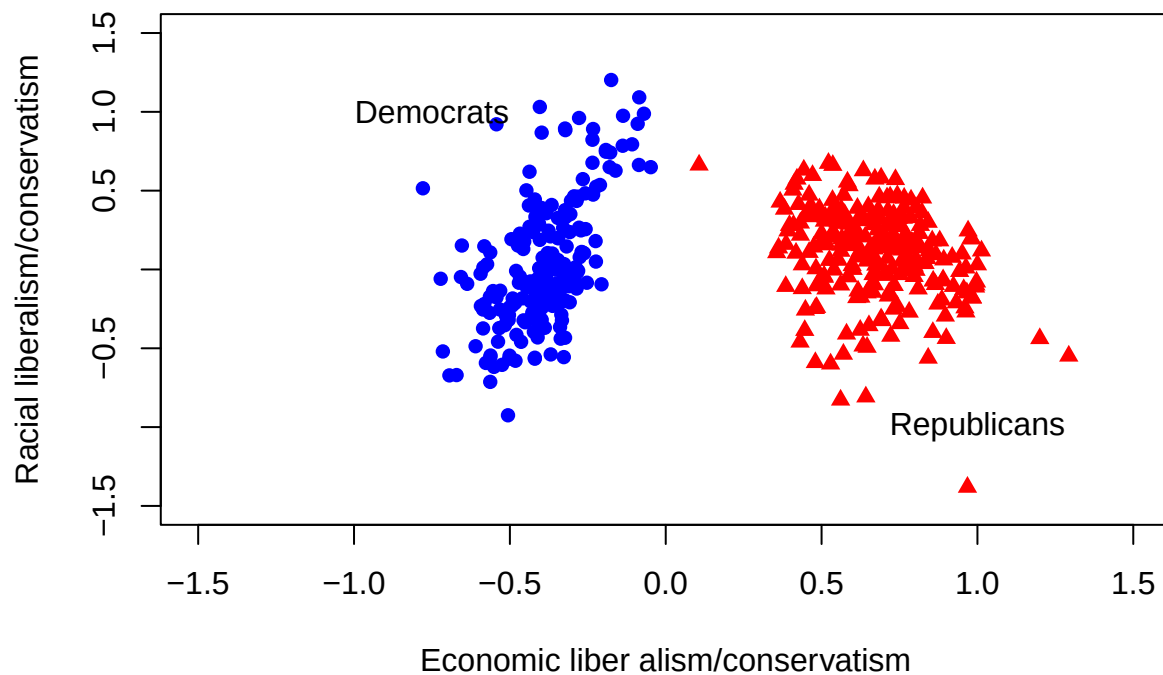


Class Exercise

Can you produce an identical plot for the 112th Congress?

What do these plots show us? Does the measure have face validity?

112th Congress



Let's Dig into this Further

How have the median R and D Members of Congress changed over time?

First calculate the median for each party by Congress:

```
dem.median <- tapply(dem$dwnom1, dem$congress, median) #Find the median of dwnom1 for each congress. Y  
rep.median <- tapply(rep$dwnom1, rep$congress, median)
```

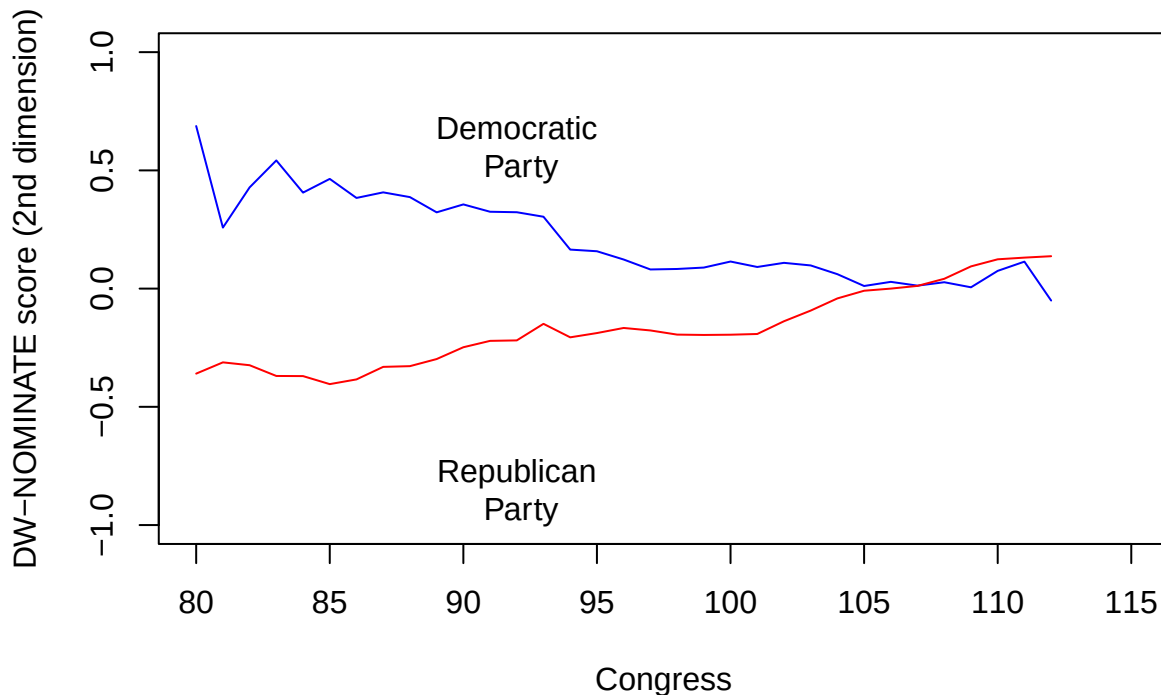
tapply is a useful command, but you won't need it for problem sets etc. Dplyr group_by does similar things

Now let's plot those medians for each party over time:

```
plot(names(dem.median), dem.median, col = "blue", type = "l", xlim = c(80, 115),  
      ylim = c(-1, 1), xlab = "Congress", ylab = "DW-NOMINATE score (1st dimension)")  
lines(names(rep.median), rep.median, col = "red")  
text(110, -0.6, "Democratic\n Party")  
text(110, 0.85, "Republican\n Party")
```

Now repeat for the 2nd Dimension

```
dem.median2 <- tapply(dem$dwnom2, dem$congress, median) #Find the median of dwnom1 for each congress.  
rep.median2 <- tapply(rep$dwnom2, rep$congress, median)  
  
plot(names(dem.median2), dem.median2, col = "blue", type = "l", xlim = c(80, 115),  
      ylim = c(-1, 1), xlab = "Congress", ylab = "DW-NOMINATE score (2nd dimension)")  
lines(names(rep.median2), rep.median2, col = "red")  
text(92, 0.6, "Democratic\n Party")  
text(92, -0.85, "Republican\n Party")
```



Bonus:

Why might we want to use the median instead of the mean here? Does using the mean change the broader story?

Really interesting visualization of DW Nominat data: <https://www.youtube.com/watch?v=M193ZIOFk0k>
Make sure to knit and, if you'd like, save the data.